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ABOUT ME

Applied Mathematics student specialized in complex systems modeling and data driven solutions. Experienced in Python and R for simulation, geospatial data analysis, statistical modeling, network analysis, and data-driven policy research.

EDUCATION

Bachelor of Science in Applied Mathematics (Expected 2026)

National Autonomous University of Mexico (UNAM)

- ✓ Relevant coursework: Mathematical Modeling, Numerical Methods, Probability, Statistics, Optimization, Complex Systems, Network Science, Machine Learning, Computational Mathematics.

EXPERIENCE

Research Assistant

Aug 2025 – Jun 2026

Center for Complexity Science (C3), UNAM

- ✓ Co-developed a scientometric network analysis software for processing scientific metadata, integrating data processing, network visualization, and configurable graph rendering into a deployable tool for qualitative scientometric analysis.
- ✓ Designed and implemented a scalable Bronze–Silver–Gold ETL pipeline processing >340M DOI records using DuckDB and Apache Parquet, with schema validation, data lineage logging, and idempotent execution for reproducibility.
- ✓ Performed large-scale statistical analysis of the CD5 disruption index across 13M scientific publications using Bayesian hierarchical models, quantile regression, and partition-based storage for memory-efficient querying.
- ✓ Developed LSTM neural networks for automated EEG artifact detection and applied the Kuramoto synchronization model to study collective neuronal dynamics in multichannel recordings.
- ✓ Designed algorithms for automated retrieval and integration of bibliographic meta-data to construct co-authorship, citation, and collaboration networks.

Research Director

Feb 2024 – Dec 2024

IMEFU MDL, Faculty of Science, UNAM

- ✓ Directed a student research division of >100 members, coordinating academic activities on economics, finance, and quantitative analysis.
- ✓ Organized seminars, workshops, and panel discussions with researchers and students nationwide; led collaborative initiatives with other student organizations.
- ✓ Authored articles on economic/financial topics for a national student publication and mentored undergraduates in research methodology and presentation skills.

SELECTED RESEARCH PROJECTS

Hack the Goal — Finalist

2025

UNESCO, Embassies of Sweden and Norway in Mexico

- ✓ Finalist in national innovation challenge to solve mobility problems during the FIFA World Cup 2026 in Mexico.

- ✓ Developed a geospatial and network -based mobility model for large - scale events, adapted to the three Mexican host cities, with direct applications to urban planning and public policy.
- ✓ Designed a business plan and implementation strategy, and presented the project before an interdisciplinary panel of experts.

PRESENTATIONS

- ✓ **Identification and Application of Indicators for Measuring the Disruptiveness of Mexican Science** (C3 UNAM, Oct 2025)
- ✓ **AI-Based Artifact Detection in EEG Signals** (C3 UNAM, Jun 2025)
- ✓ **What is Model United Nations and Why Does It Matter?** (IMEF Workshop, Jul 2025)
- ✓ **How to Become a Successful MUN Delegate** (IMEF Workshop, Jul 2025)

TECHNICAL SKILLS

- **Programming Languages** Python, R, SQL (PostgreSQL)
- **Scientific Computing** NumPy, SciPy, Pandas, Matplotlib, Jupyter
- **Machine Learning** TensorFlow, Keras, Optuna, LSTM Neural Networks
- **Data Engineering** DuckDB, Apache Parquet, ETL Pipeline Design, Data Validation, Schema Design, Metadata Processing
- **Mathematical and Statistical Methods** Bayesian Inference, Generalized Linear Models, Monte Carlo Simulation, Bootstrap, Optimization, Complex Systems Modeling, Network Science, Time Series Analysis, Numerical Simulation
- **Geospatial Analytics** OpenStreetMap Data Processing, Urban Mobility Modeling, Spatial Networks, Geospatial Data Analysis
- **Software and Tools** Git, GitHub, Linux, L^AT_EX

RESEARCH INTERESTS

Complex Systems, Network Science, Machine Learning, Scientific Computing, Bayesian Statistics, Scientometrics, Computational Social Science, Urban Analytics, Public Policy.